SARAH D. BUSSLINGER

sarah.busslinger@psi.ch

+41796750675

5400, Baden Switzerland

EDUCATION

Ph.D.:
Pharmaceutical Sciences
ETH Zurich, Switzerland,
2020 - Current

Master of Science:
Biochemistry
University of Zurich,
Switzerland, 2017 - 2019
"Generation of half-life optimized polypeptide-DARPin-MMAF conjugates for targeted tumor therapy"

Exchange Year:
Swiss European Mobility Programme (SEMP)
University College Dublin,
Ireland, 2016 - 2017

Bachelor of Science:
Biology University of Zurich, Switzerland, 2014 - 2017

PROFESSIONAL SUMMARY

I am currently a Ph.D. student working in the group of PD Dr. Cristina Müller and Prof. Dr. Roger Schibli at the Center for Radiopharmaceutical Sciences at PSI, Switzerland, focusing on the development of novel tumor-targeted radionuclide therapies. I studied at the University of Zurich (Switzerland) from which I received my Bachelor of Science in Biology in 2017 and graduated my Master of Science in Biochemistry in 2019. During my Master's Thesis I worked on the "*Generation of Half-Life Optimized Polypeptide-DARPin-MMAF Conjugates for Targeted Tumor Therapy*" in the group of Prof. Dr. Andreas Plückthun.

SKILLS

- Cancer research
- Radionuclide therapy
- Protein-drug conjugates
- Mammalian cell culture
- Cell-based in vitro assays
- Radiolabeling
- Molecular cloning
- PCR
- Protein expression and purification
- HPLC and FPLC
- In vivo investigation of tumor-targeting agents (Biodistribution, SPECT/CT, tumor therapy)

WORK HISTORY

April 2020 - Current
Paul Scherrer Institute - PhD Student, Villigen-PSI
- Development of novel tumor-targeted radioligand therapies

March 2019 - February 2020
University of Zurich - Research Associate, Zurich
- Optimizing the anti-tumor efficacy of DARPin-drug conjugates by engineering the molecular size and half-life

LANGUAGES

German

Mother Tongue (C2)

English

Bilingual or Proficient (C2)

August 2010 - September 2016

Student Jobs – Tutoring

- Teaching different students from 6th grade up to Bachelor of Science students mathematics, biology and chemistry

---

CERTIFICATIONS

- Swiss labor animal research certification (LTK1 & LTK2)

PUBLICATIONS

Busslinger SD, Tschan VJ, Richard OK, Talip Z, Schibli R, Müller C. *[^{225}Ac]Ac-SibuDAB for Targeted Alpha Therapy of Prostate Cancer: Preclinical Evaluation and Comparison with [^{225}Ac]Ac-PSMA-617*. Cancers (Basel). 2022; 14(22):5651. doi: 10.3390/cancers14225651

Co-authorship:

Tschan VJ, Borgna F, Busslinger SD, Stirn M, Rodriguez JMM, Bernhardt P, Schibli R, Müller C. *Preclinical investigations using [^{177}Lu]Lu-Ibu-DAB-PSMA toward its clinical translation for radioligand therapy of prostate cancer*. Eur J Nucl Med Mol Imaging. 2022; 49(11):3639. doi: 10.1007/s00259-022-05837-2

Borgna F, Deberle LM, Busslinger SD, Tschan VJ, Walde LM, Becker AE, Schibli R, Müller C. *Preclinical Investigations to Explore the Difference between the Diastereomers [^{177}Lu]Lu-SibuDAB and [^{177}Lu]Lu-RibuDAB toward Prostate Cancer Therapy*. Mol Pharm. 2022; 19(7):2105. doi: 10.1021/acs.molpharmaceut.1c00994

Benešová M, Guzik P, Deberle LM, Busslinger SD, Landolt T, Schibli R, Müller C. *Design and Evaluation of Novel Albumin-Binding Folate Radioconjugates: Systematic Approach of Varying the Linker Entities*. Mol Pharm. 2022; 19(3):963.
doi: 10.1021/acs.molpharmaceut.1c00932

Brandl F, Busslinger S, Zangemeister-Wittke U, Plückthun A. *Optimizing the anti-tumor efficacy of protein-drug conjugates by engineering the molecular size and half-life*. J Control Release. 2020; 327:186. doi: 10.1016/j.jconrel.2020.08.004

PRESENTATIONS AT NATIONAL AND INTERNATIONAL CONFERENCES

International Symposium on Radiopharmaceutical Sciences (2022)
Novel Folate Radioconjugates: Optimization of Functional Units to Improve the Tissue Distribution Profile

Swiss Congress of Radiology (2022)
Novel Folate Radioconjugates: Optimum Combination of Functional Units to Improve the Tissue Distribution Profile

78. Jahresversammlung Schweizerische Gesellschaft für Urologie (2022)
Preclinical Evaluation of [^{225}Ac]Ac-SibuDAB for Targeted alpha Therapy of Prostate Cancer

PERSONAL INTERSTS

- Sports / Fitness
- Skiing / Snowboarding / Freeriding
- Bouldering
- Traveling